

FAST-NUCES Karachi (Main Campus)
Database Management Systems - Project Proposal
BSCS 4D

Hospital Management System (HMS)

Group Members

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1. Project Overview

The Hospital Management System (HMS) is a full-stack web application designed to digitize and manage the core operational workflows of a hospital environment. It brings together four distinct user roles—Admin, Doctor, Receptionist, and Patient—under one unified platform, each with precisely scoped access rights, a dedicated dashboard, and a defined set of permitted actions. The system is underpinned by a relational PostgreSQL database that enforces referential integrity, domain constraints, and a carefully designed Role-Based Access Control (RBAC) model.

2. Project Objectives

- Provide a centralized, role-aware digital platform for managing hospital operations.
- Enforce data integrity through primary keys, foreign keys, unique constraints, and check constraints.
- Implement a full appointment lifecycle with deterministic status transitions.
- Record clinical outcomes (interactions, diagnoses, prescriptions) atomically using database transactions.
- Deliver real-time, role-specific dashboard statistics using aggregation queries.
- Protect sensitive patient data through JWT-based authentication and RBAC permission checks.

3. Proposed Technology Stack

- Frontend & API Layer: Next.js 15, React, TypeScript, Tailwind CSS
- Database: PostgreSQL 15+ (sole source of truth)
- DB Driver: node-postgres (pg)
- Auth & Middleware: JWT (jsonwebtoken), bcryptjs, Next.js Middleware

4. Database Schema Overview

The database will consist of 13 tables divided into two logical groups, adhering strictly to the Third Normal Form (3NF):

4.1 RBAC Infrastructure Tables

Implements the NIST RBAC Standard Level 1 model to manage permissions.

- Users: Central identity store. Primary keys use UUIDs generated via pgcrypto.
- Role: Defines finite system roles (admin, doctor, receptionist, patient).
- Permission: Atomic action descriptors.
- UserRole & RolePermission: Junction tables enforcing many-to-many mappings.

4.2 Domain Tables

Models the operational entities connected via a hub-and-spoke model to the Users table.

- Patient: Demographic and profile data.
- Doctor: Professional specialization and availability.
- Receptionist: Staff profile.
- Appointment: The central operational table tracking scheduled visits.
- Interaction, Diagnosis, Prescription: Tables modeling the hierarchical clinical record.

5. Key DBMS Concepts to Implement

- Data Integrity Constraints: Comprehensive use of UNIQUE constraints (e.g., email), BIGSERIAL sequences, and ON DELETE CASCADE/SET NULL tailored to relationship semantics.
- Advanced Querying: Implementation of complex joins (INNER/LEFT), conditional COUNT aggregates with FILTER clauses, and json_agg functions for deep relational data retrieval.
- ACID Transactions: Usage of BEGIN, COMMIT, and ROLLBACK to guarantee the atomic insertion of clinical data across multiple tables simultaneously.
- Database Triggers: Implementation of a BEFORE UPDATE trigger function to automatically maintain timestamps (updatedAt) across user-editable tables independent of the application layer.
- Security: Parameterized queries throughout to prevent SQL injection, coupled with robust connection pooling.